User Manual –

PureStream

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Introduction

PureStream is a Java Swing–based desktop application designed to provide users with an easy, ad-free solution for downloading, converting, and managing multimedia content through a modern and intuitive graphical interface.

The application integrates powerful command-line tools such as yt-dlp for downloading online media and FFmpeg for post-processing and format conversion, while completely hiding their complexity behind a user-friendly desktop interface. This allows users to perform advanced multimedia tasks without requiring technical knowledge of terminal commands.

In addition to media downloading, PureStream includes a media management system that lets users browse, filter, and delete downloaded files directly from the application. A built-in login system ensures controlled access, with support for persistent sessions and secure logout.

Intended Audience
PureStream is intended for:

Beginner users who want to download video or audio content easily without dealing with command-line tools or complex configurations.
Intermediate users who need additional control over formats, file management, and download tracking while maintaining a simple workflow.
Users who value usability and visual feedback, benefiting from real-time progress tracking, clear error messages, and responsive UI behavior.
Anyone who wants an efficient desktop tool to download and organize multimedia content, customize the interface with light or dark mode, and manage files from a single application.
The main goal of PureStream is to demonstrate how a well-designed desktop application can combine powerful background processing, clean user experience (UX), and modular architecture into a cohesive and accessible product.

System Requirements

To ensure proper operation of PureStream, the following system requirements must be met. The application is designed to be lightweight and compatible with most systems.

Operating System

PureStream is a cross-platform desktop application and can run on:

Windows 10 or newer

- Linux
    - Ubuntu
    - Debian-based distributions
- macOS (with Java properly installed)

Java Environment

- Java Development Kit (JDK) 17 or newer
- Java must be properly installed and available in the system PATH

<u>To verify installation:</u>

- **java -version**

Required External Tools

To ensure proper operation of PureStream, the following tools and requirements must be available on your system.

yt-dlp

Used for downloading video and audio content.

- Must be installed on your system
- Must be accessible from the system PATH

<u>Check installation:</u>

yt-dlp --version

FFmpeg

Used for audio and video conversion and post-processing.

Required for:

- MP3 extraction
- MP4 conversion

<u>Check installation:</u>

ffmpeg -version

Disk Space

- Minimum 200MB of free disk space for the application
- Additional free space required to download videos and audio files

Internet Connection

<u>Required for:</u>

- User authentication
- Downloading media content

Dependencies

.jar file from MediaPollingClientComponent
Installation

Installation of the Application

Step 1 → Clone the repository
Open a terminal and run the command:

```
git clone https://github.com/RominaRodriguez5/Pure-Stream.git
```

Step 2 → Build the Project
Run the following command to build the project:

```
mvn clean package
```

Step 3 → Run the Application
Use the following command to open the application:

```
java -jar target/PureStream.jar
```

Installation of the Component (Required)

Option A → Using an IDE

Step 1 → Open the project in your IDE (NetBeans / IntelliJ / Eclipse)

- Application:
  https://github.com/RominaRodriguez5/Pure-Stream.git

- Component:
  https://github.com/RominaRodriguez5/MediaPollingClientComponent.git

Step 2 → Add the Component to the Project
Add MediaPollingClientComponent.jar inside the PureStream project.

Step 3 → Rebuild the Project
Rebuild the project to apply the changes.

Option B → Manual Setup

Step 1 → Create the libs Folder
In the application project root, create a folder called: libs/

Step 2 → Add the Component
Place MediaPollingClientComponent.jar inside the libs/ folder.

Step 3 → Run the Application
Run the application with the component included.

First Launch

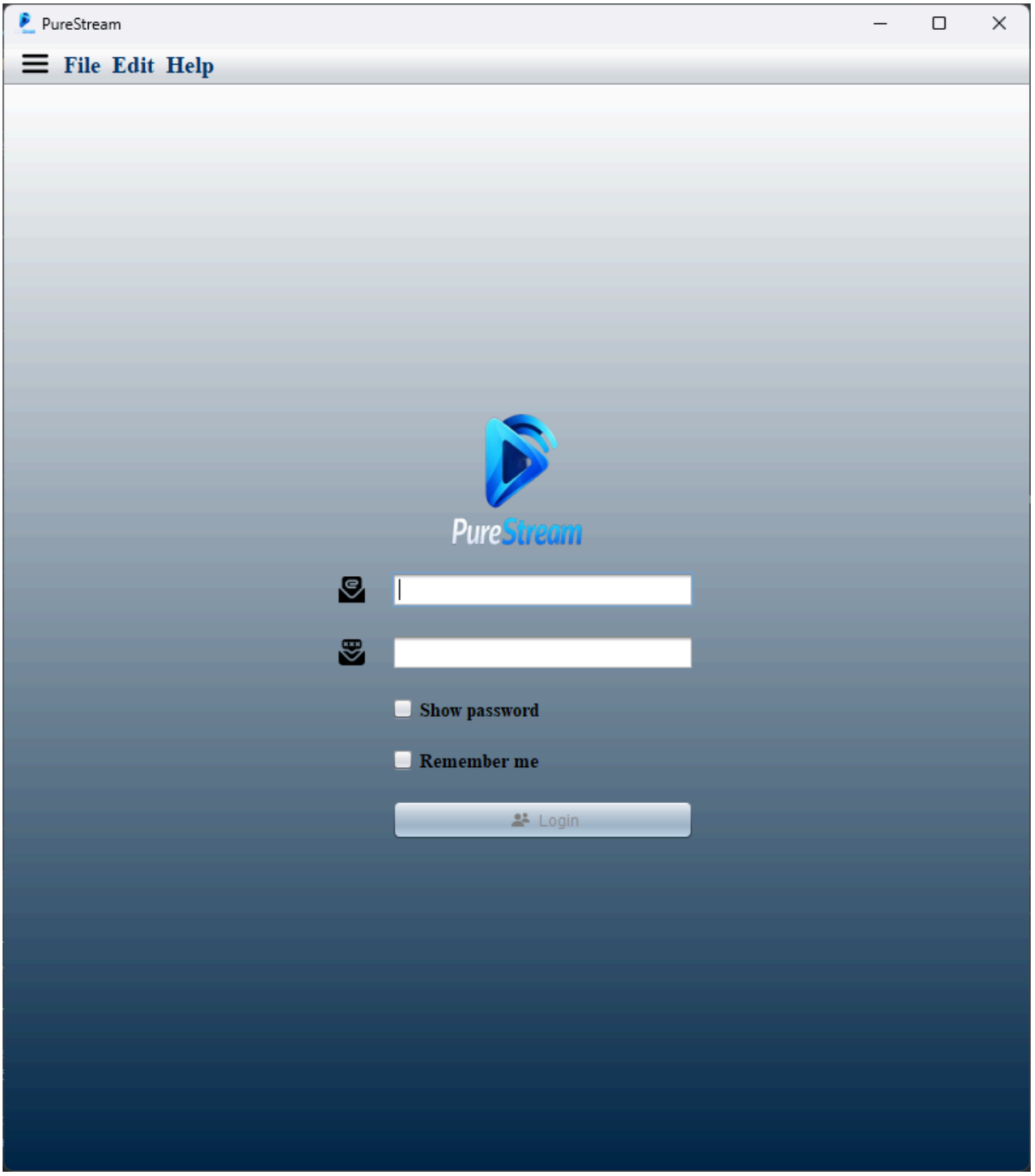
When you open PureStream for the first time, the application will guide you through a simple and intuitive startup process.

What Happens on First Launch

- The Login screen appears automatically.
- The application opens the Login Panel.
- You must authenticate before accessing the application.

User Authentication

- Enter your email and password.
- When both fields are filled, the Login button becomes enabled.
- There is a checkbox to show the password you entered (optional).
- Optionally enable Remember Me to save your credentials.

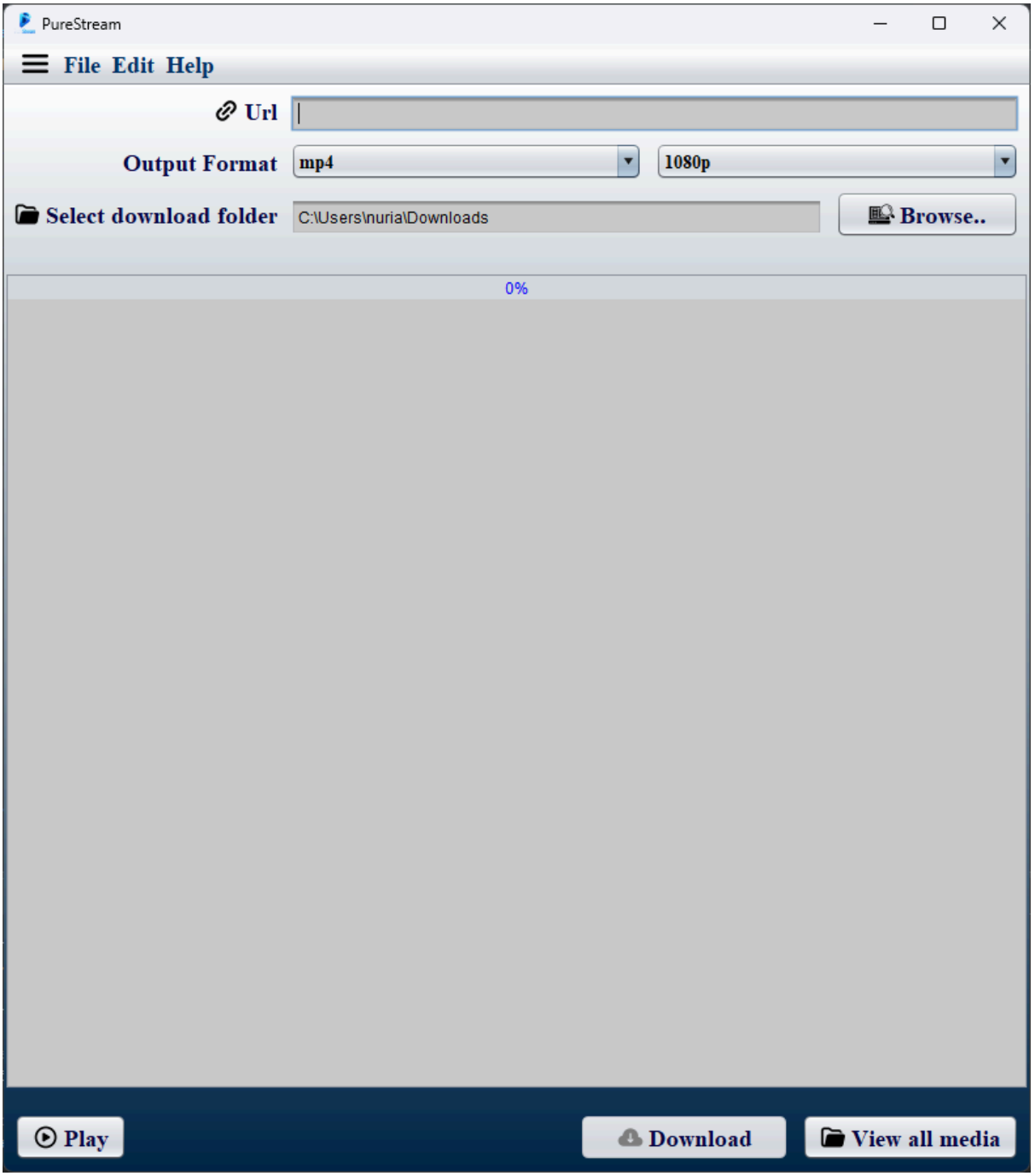


Interfaces

Main Frame

- URL: Add the link you want to download.
- Output Format: Choose between mp3 (audio) or mp4 (video), and select the quality (360, 480, 720, 1080).
- Select Download Folder: Click Browse to choose the path where the files will be downloaded.
- Play: Once the video is downloaded, you can play it.
- Download: Start downloading the selected video.

- View All Media: Opens the panel where you can see all downloaded media.
- Menu Bar:
    - File: Exit and Logout.
    - Edit: Preferences.
    - Help: About.

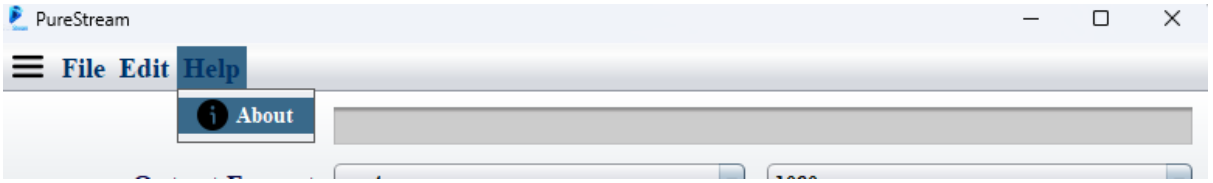


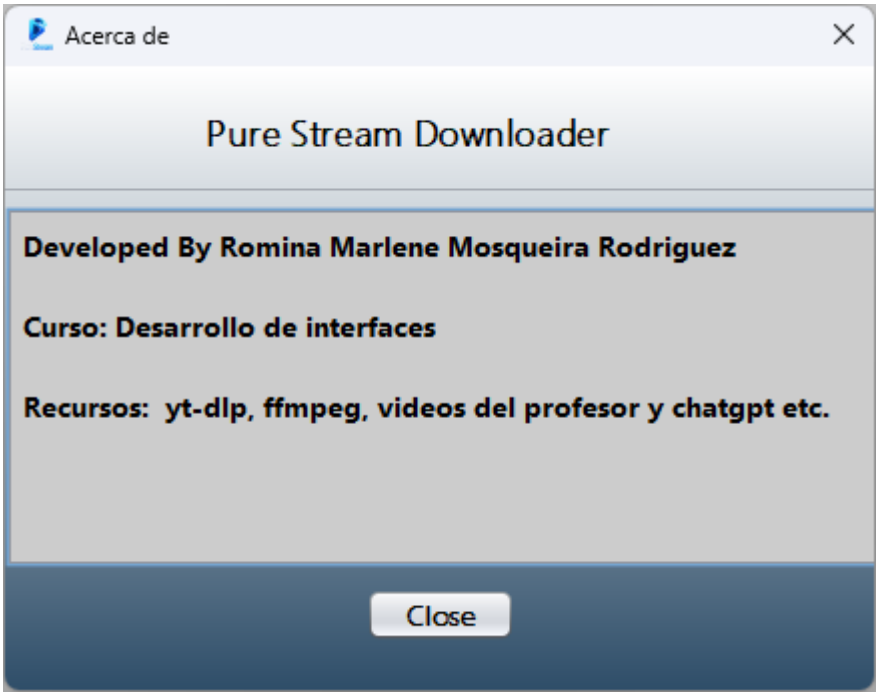
About Panel

In this panel, you can see information about:

- The application's name
- The developer

- The course
- Resources





Preferences Panel

In this panel, you can configure the following options:

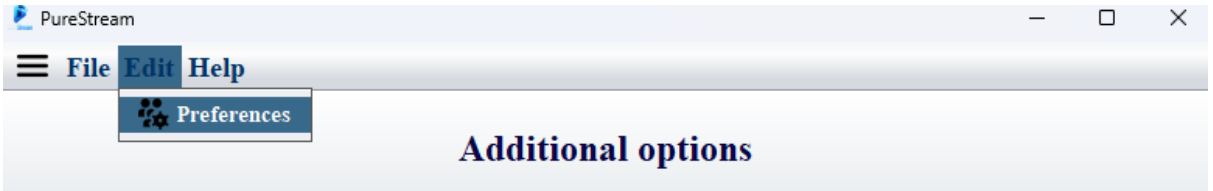
- IMPORTANT: Select the path of the yt-dlp executable.
- If this is not configured, you will not be able to download content.

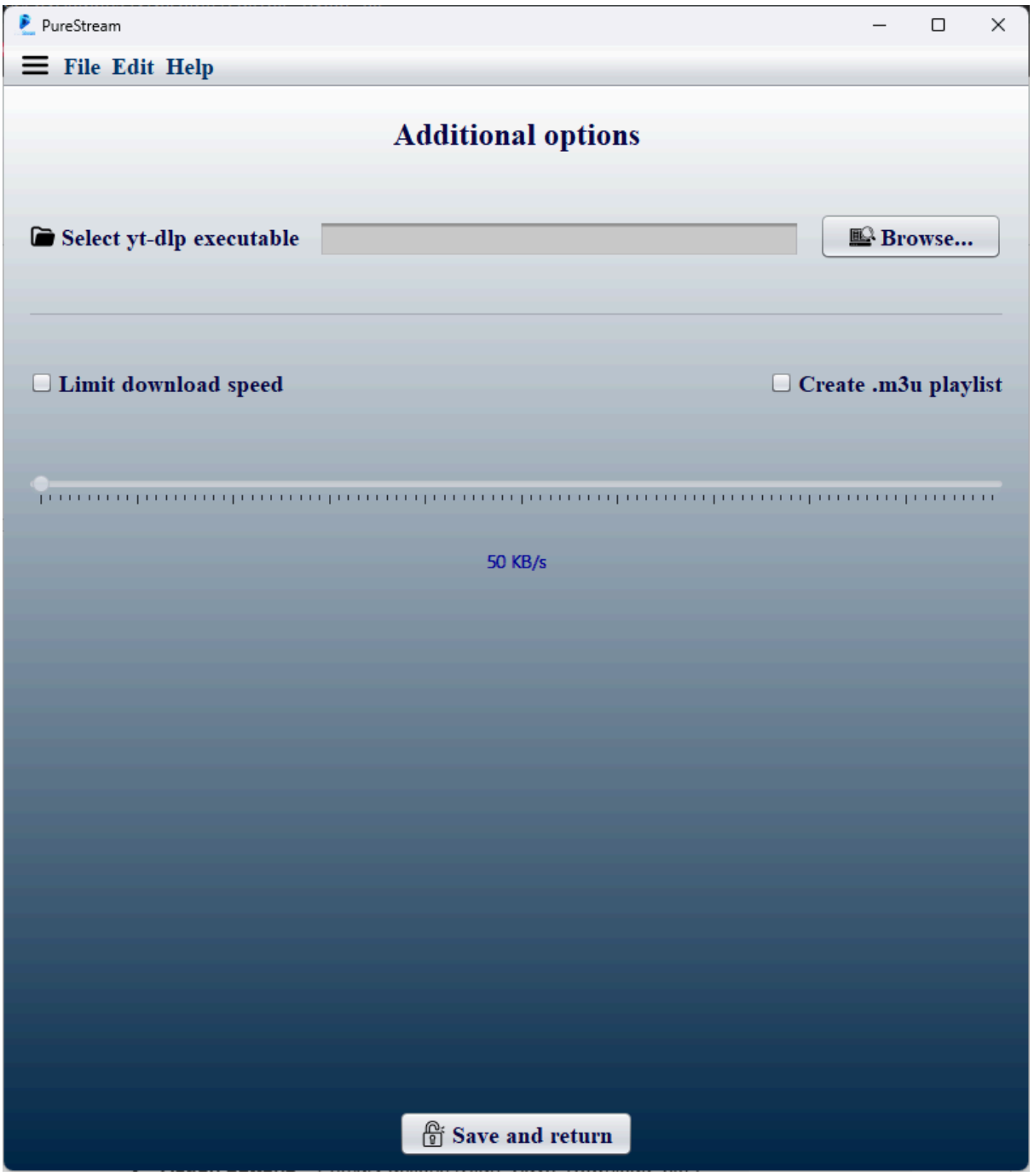
- Browse: Click this button to search for and select the yt-dlp executable.

- Speed: Enable the speed limit checkbox and choose the download limit (kb/s) using the slider.

- Playlist: Enable this checkbox to automatically create a playlist with the downloaded media.

- Save and Return: Saves your preferences and returns to the main frame.





Details Panel

In this part of the application, you can find:

Media Locations:
Three different folders where the media can be stored:

- Local
- Network
- Both

You can also see the media details: name, type, size, and date.

Bottom Panel:
Displays the selected media with a Play button on the right side.

Delete:
Deletes the selected media.

Upload to Cloud:
Uploads the selected media to the cloud.

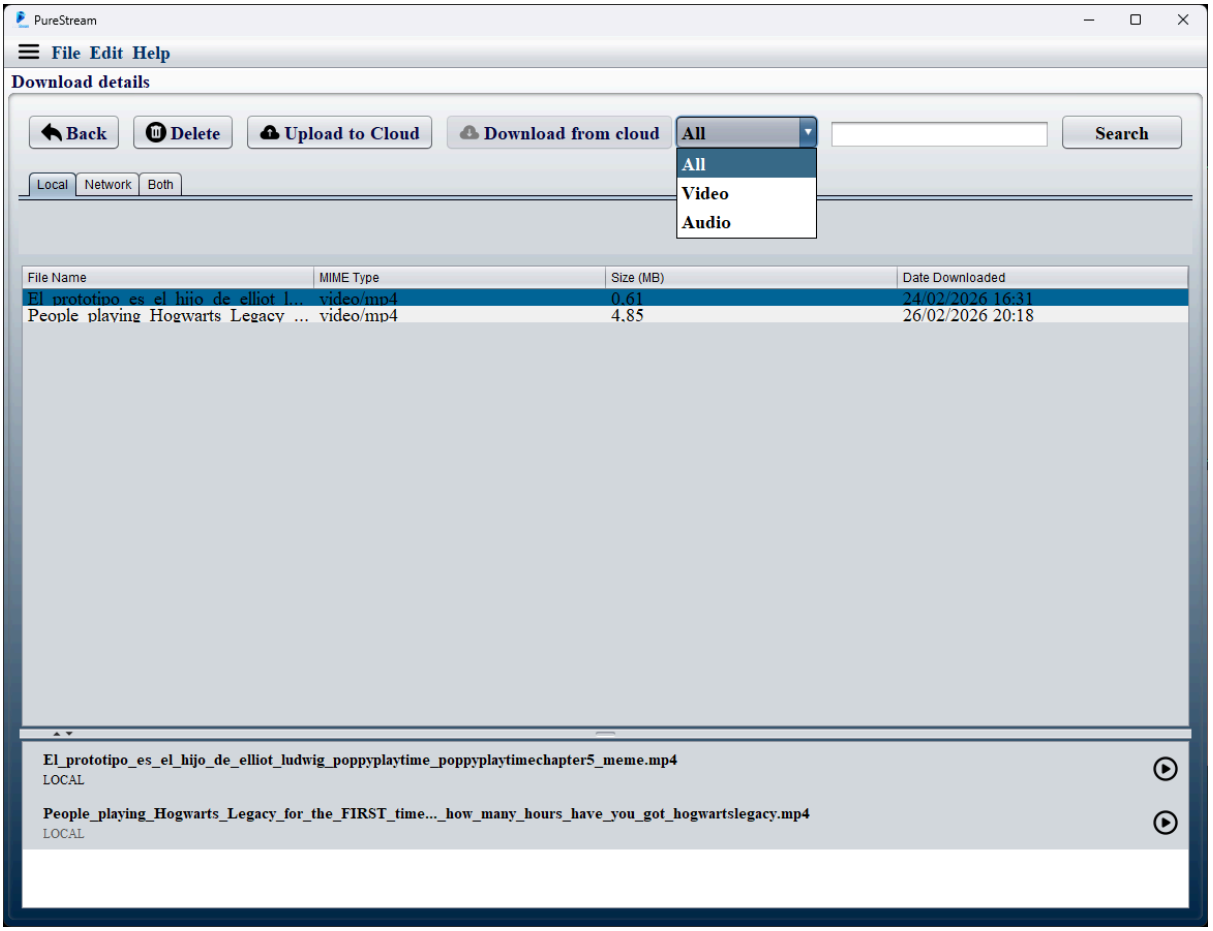
Download from Cloud:
Downloads the selected media to local storage.

Media Filter List:
Allows you to view:

- All media
- Only video
- Only audio

Search Field:
Enter the name of the media you are looking for and press the Search button.

Back Button:
Returns to the main frame.



Exit Options

The application provides two ways to leave:

Option A → Exit Completely

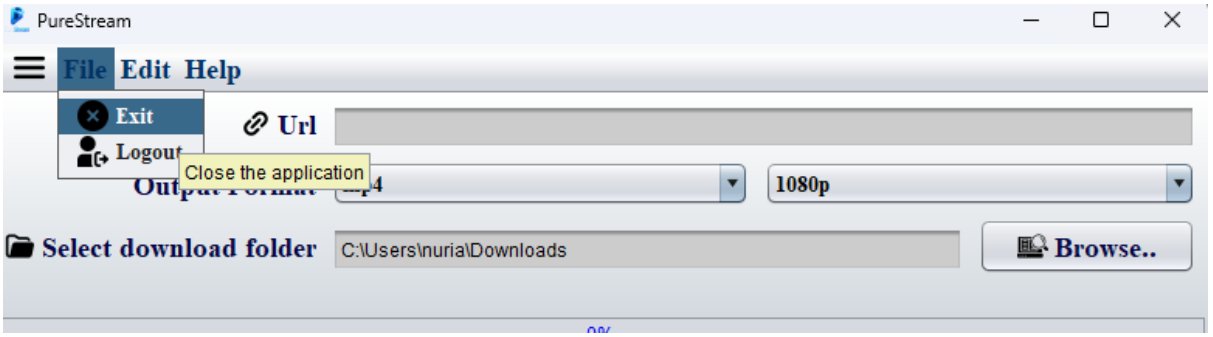
Closes the application entirely.
Go to File → Exit.
A confirmation panel will appear:

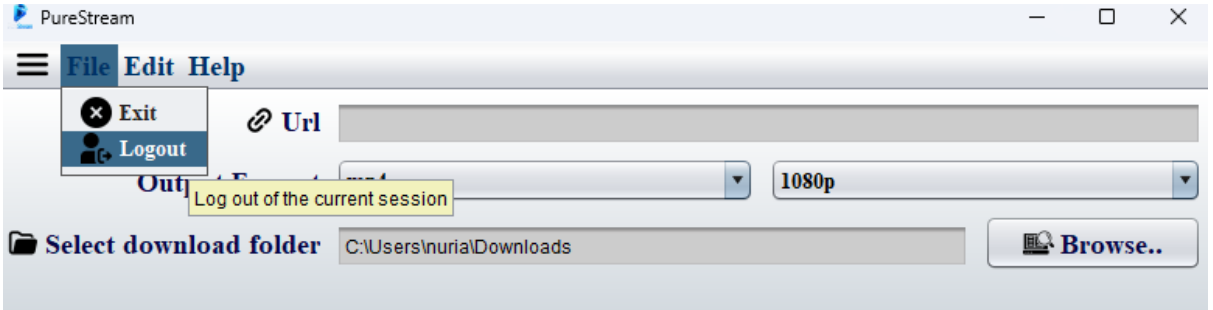
- Click Yes to close the application.
- Click No to return.

Option B → Logout

Closes your session but keeps the application open.
Go to File → Logout.
A confirmation panel will appear:

- Click Yes to log out.
- Click No to return.

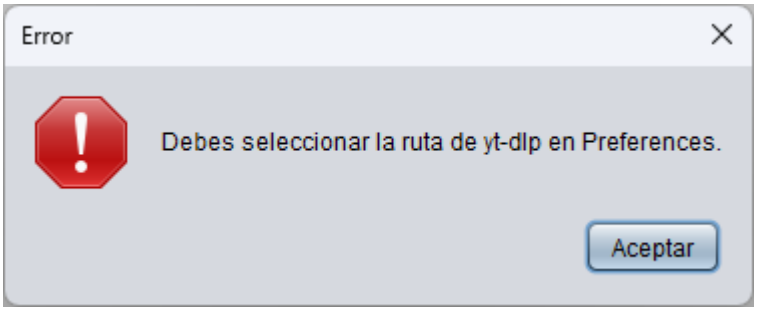




Error Messages and Troubleshooting

How Errors Are Communicated

- Dialog windows with descriptive error messages.
- Disabled buttons to prevent invalid actions.
- Visual feedback, such as progress indicators.



Troubleshooting Tips

- Make sure yt-dlp and FFmpeg are installed on your system.
- Verify your internet connection if downloads fail.
- Restart the application if a previous session was interrupted.
- If an error persists, check the console logs or open an issue on GitHub.

Support and Contact

If you encounter issues with the application or need help, follow these steps:

Open an Issue on GitHub

- Describe what you were doing.
- Explain what happened.
- Attach screenshots if possible.

Contact the Developer

- Email: rominamarlenemosqueira@paucasesnovescifp.cat